

NRLC-09: Toolchain Reference & End-to-End Runbook

Series: NRLC — New Relic to Dynatrace Migration Deep Dives | Notebook: 9 of 9 | Created: April 2026 | Last Updated: 10/05/2026

Overview

The master reference for the open-source toolchain that powers every preceding notebook in this series. This deep dive covers how `Dynatrace-NewRelic`, `nrql-engine`, and `nrql-translator` fit together, how to operate the orchestrating CLI, the configuration surface, the export formats (Monaco v2 YAML, Terraform HCL), and a complete end-to-end runbook.

As of **2026-04-15** the Python orchestrator ships **40+ transformers** across 9 phase batches (Phase 11 foundation + Phases 16–24 expansion), **1,200+ unit tests**, and **292 NRQL translation patterns** pinned to the TypeScript `nrql-engine` via the Phase 19b regression suite. It emits **Gen3 Dynatrace by default** (Workflows, anomaly detectors, Segments, OpenPipeline, Document API); use `--legacy` only for the 8 Gen2-only capabilities or for tenants without Gen3 features.

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Prerequisites

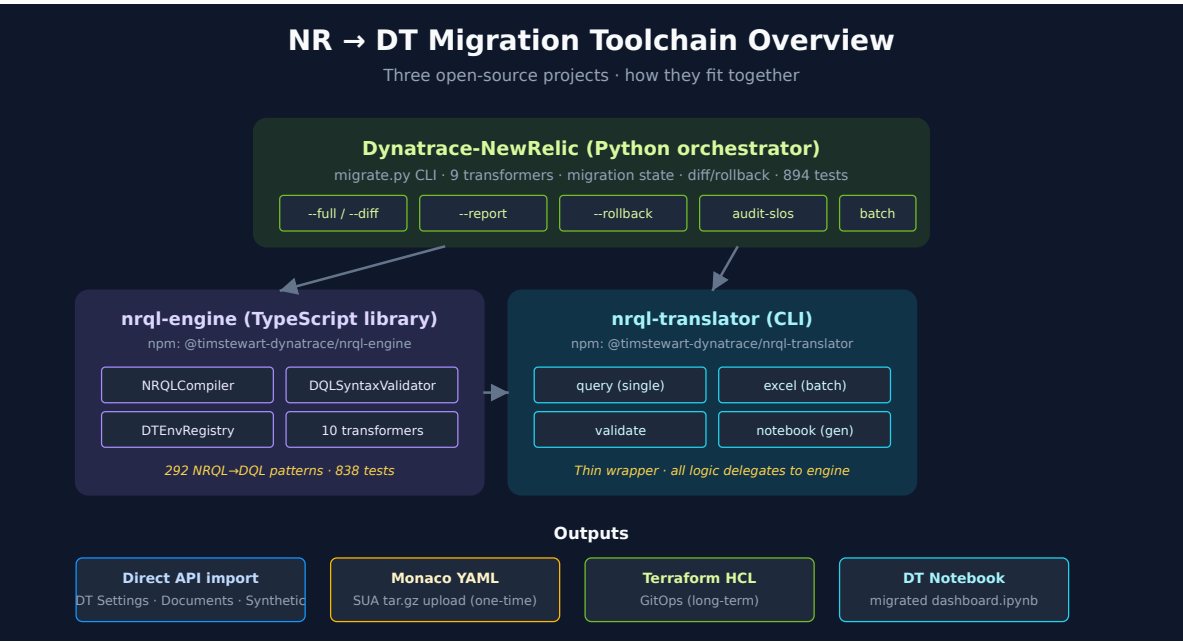
Requirement	Details
Audience	Engineers operating the migration end-to-end
Required reading	All preceding NRLC notebooks (this is the operational reference)
Access	NR API key + DT API token; both with appropriate scopes (see §2)

For the full Terraform resource catalog (worked examples for every resource named in §5's export list, plus the OAuth-client-only vs. Platform-Token distinction for IAM/bucket resources), see **AUTOM-04** (Terraform Provider) and **AUTOM-02** (Settings API) rather than treating this notebook as the canonical shape.

1. The Three Projects — How They Fit



Use Case	Tool
End-to-end NR → DT migration	Dynatrace-NewRelic (migrate.py)
Embed translation in your own tool/web app	nrql-engine
One-off NRQL → DQL on the command line	nrql-translator
Translate a batch of dashboard queries	nrql-translator excel or migrate.py batch
Generate a DT notebook from translated queries	nrql-translator notebook



Upstream Home: Dynatrace Migration Assistant (dynatrace-dma)

The official Dynatrace migration tooling lives under the [dynatrace-dma](#) GitHub organization — the **Dynatrace Migration Assistant** project. Existing repos in the org:

Repo	Source Platform	Status
splunk-to-dynatrace	Splunk	Active
datadog-to-dynatrace	Datadog	Active
newrelic-to-dynatrace <i>(planned)</i>	New Relic	nrql-engine planned to relocate here

When the New Relic engine relocates, the recommended install path will become the `dynatrace-dma` repo. Until then, `timstewart-dynatrace/nrql-engine` is canonical.

2. Installation & Configuration

Dynatrace-NewRelic (Python)

```
git clone https://github.com/timstewart-dynatrace/NewRelic-to-Dynatrace-Migration-Utilities
cd NewRelic-to-Dynatrace-Migration-Utilities
pip install -r requirements.txt
cp .env.example .env
# Edit .env with NR + DT credentials
```

Future home note: The `nrql-engine` underlying this orchestrator is planned to relocate to the `dynatrace-dma` (Dynatrace Migration Assistant) organization, joining the existing `splunk-to-dynatrace` and `datadog-to-dynatrace` repos. Track that org for the official Dynatrace-supported install going forward.

Required NR API Token Scopes

- `read:dashboards`, `read:alerts`, `read:nrql`, `read:synthetics`, `read:slo`, `read:workloads`, `read:logs`

Required DT API Token Scopes

- `dashboards.read` + `dashboards.write` (or Documents v2: `documents:read`, `documents:write`)
- `metrics.read`
- `entities.read`
- `settings:objects.read` + `settings:objects.write`
- `slo.read` + `slo.write`
- `synthetic.read` + `synthetic.write`
- `notifications.write`
- `iam-policies.read` (for IAM policy + OpenPipeline configuration validation)

Note: These are Classic API token scopes (the `<area>.<action>` style). **Platform Tokens use policy-based permissions** instead and are the recommended default on Gen3 tenants — see Dynatrace IAM docs for the equivalent policy statements. OAuth clients follow the Classic scope style for programmatic access.

Configuration File (`.env`)

Key names as in the repository's `.env.example` (main branch, read 10/05/2026):

```

NEW_RELIC_API_KEY=NRAK-...
NEW_RELIC_ACCOUNT_ID=1234567
NEW_RELIC_REGION=US          # or EU

DYNATRACE_API_TOKEN=dt0c01.XXXX...
DYNATRACE_ENVIRONMENT_URL=https://abc12345.live.dynatrace.com
DYNATRACE_DETECTOR_ACTOR=    # service-user UUID; anomaly detectors fail to import without it

# Optional
MIGRATION_DRY_RUN=false
MIGRATION_OUTPUT_DIR=./output
LOG_LEVEL=INFO

```

`audit-slos` additionally reads `DYNATRACE_OAUTH_TOKEN` from the environment and exits without it.

Settings are loaded via Pydantic; missing required values fail fast with descriptive errors.

3. `migrate.py` CLI Reference

Subcommand inventory (post-Phase-24)

The `migrate.py` entry point registers **14 subcommands** (`main` branch, read 10/05/2026):

`agents`, `archive`, `audit`, `audit-slos`, `batch`, `compile`, `convert`, `export-monaco`, `export-terraform`, `extract-nrql`, `migrate`, `preflight`, `reference`, `scan-instrumentation`

Subcommand	Purpose
<code>migrate</code>	Full pipeline (export → transform → import). Accepts <code>--full</code> , <code>--export-only</code> , <code>--import-only</code> , <code>--dry-run</code> , <code>--components</code> , <code>--rollback <file></code> , <code>--retry <file></code> , <code>--resume</code> , <code>--incremental</code> , <code>--report</code> , <code>--diff</code> , <code>--legacy</code> , <code>--canary <pct></code> , <code>--canary-auto-proceed</code>
<code>compile</code>	Single NRQL → DQL compilation. Flags: <code>--interactive</code> , <code>--file</code> , <code>--validate</code> , <code>--output</code>
<code>convert</code>	NRQL → DQL with post-processing + auto-fix (wraps <code>compile</code> + <code>DQLFixer</code>)
<code>batch</code>	CSV / Excel batch compile (NRQL column)
<code>reference</code>	Print NRQL → DQL reference table; <code>--mappings</code> for full mapping tables (230 metrics + 72 attributes + 90+ aggregations + 34 event types)
<code>preflight</code>	Phase 14. Probe target DT tenant for Gen3 API availability (Settings 2.0 / Document / Automation). Suggests <code>--legacy</code> if any surface is missing.
<code>agents</code>	Phase 16. Per-language APM agent migration action plans (Java / .NET / Node.js / Python / Ruby / PHP / Go). Flags: <code>-language <lang></code> , <code>--phase</code> , <code>--dry-run</code>
<code>scan-instrumentation</code>	Phase 16. Scan one source file (<code>--file</code>) for <code>newrelic.*()</code> SDK calls; print DT/OTel replacement suggestions as a diff (<code>--output</code> writes it to a file). Side-effect-free; manual apply.
<code>archive</code>	Phase 17. Pre-decommission NRDB snapshot (resumable JSONL per event type). Requires <code>--account</code> and <code>--since</code>
<code>audit</code>	Phase 20. Config drift: compares a transformed baseline (<code>--baseline <dynatrace_config.json></code>) with the live tenant. Reports RENAMED / DELETED / MODIFIED / EXTRA. Read-only; exits 1 on drift. Does not query New Relic.
<code>audit-slos</code>	Validate DT SLOs against live metrics for missing/invalid keys (Phase 11 + Phase 20 enrichment). DT side only; needs <code>DYNATRACE_OAUTH_TOKEN</code>
<code>extract-nrql</code>	Extract every NRQL query from the export JSON into a flat <code>.txt</code> or <code>.csv</code> list.
<code>export-monaco</code>	Emit Monaco v2 project YAML (Gen3 default; <code>--legacy</code> for Gen2 shapes).
<code>export-terraform</code>	Emit Terraform HCL with <code>dynatrace-oss/dynatrace</code> provider (Gen3 default; <code>--legacy</code> for Gen2 shapes).

Note: No subcommand runs NRQL against New Relic and DQL against Dynatrace to compare results. Behavioral validation

(NRCL-08 §4) is a manual step; `audit` compares *configuration*, not query results.

migrate flags

These are options of the `migrate` subcommand, not global flags. There is no `--transform-only`, `--filter`, `--log-level` or `--no-color`.

Flag	Purpose
<code>--components <list></code>	Restrict to specific components (see §4 for the accepted names)
<code>--export-only</code> / <code>--import-only</code>	Run one phase only; <code>--import-only</code> needs <code>--input <dir></code> and writes no rollback manifest
<code>--dry-run</code>	Export and transform, skip the import ("Validate without applying changes")
<code>--diff</code>	Compare the transformed entities with the live tenant — use with <code>--dry-run</code> , otherwise it runs after the import
<code>--output <dir></code>	Override output directory
<code>--rollback <file></code>	Delete the entities listed in a rollback manifest
<code>--legacy</code>	Gen2 output path — emits Alerting Profiles / Management Zones / Auto-Tags / Config v1 dashboards instead of Gen3 equivalents.
<code>--canary <pct></code>	Phase 20. Two-wave import — import N% first, await approval, then import remainder
<code>--canary-auto-proceed</code>	In canary mode, skip the interactive prompt and proceed automatically (CI / scripted use)
<code>--report</code>	Emit enriched conversion report (<code>confidence_score</code> , <code>warning_codes</code> , <code>runbook_url</code> per entry)

Transformer coverage

`migrate.py` orchestrates 40+ entity transformers across 9 phase batches. See the [coverage matrix \(migration utilities GitHub\)](#) for the complete NR-surface → transformer mapping. High-level summary:

Phase	Transformers added
11 (foundation)	alert, dashboard, synthetic, slo, workload, infrastructure, log_parsing, tag, drop_rule
16 (P0 coverage)	agents/ orchestrators (7 languages), lambda, browser_run, mobile_run, custom_instrumentation_translator
17 (P1 alerts + data + identity)	non_nrql_alert, baseline_alert, lookup_table, maintenance_window, change_tracking, custom_event_ingest, identity, log_obfuscation, tools/nrdb_archive
18 (specialized products)	cloud_integration, kubernetes, aiops, vulnerability, npm, ai_monitoring, prometheus
19 (dashboard widget parity)	funnel / honeycomb / event-feed / cascading vars / saved views / permissions uplift
19b (nrql-engine parity)	compiler/shorthands, K8s metric + entity-field overrides, 24 DQL fixer methods
20 (operational safety)	migration/canary, migration/audit, DynatraceClient.delete_entity, enriched ConversionReport
23 (second-wave parity)	key_transaction, otel_metrics, statsd, cloudwatch_metric_streams, metric_transform plugin, transformers/mappings/
24 (third-wave parity)	database_monitoring (10 engines), on_host_integration (12 techs), security_signals, custom_entity, log_archive, metric_normalization, synthetic_specialized (cert-check + broken-links), saved_filter_notebook, otel_collector, legacy/error_inbox_v1, legacy/request_naming_v1

4. Component Selection

Migrating one entity type at a time is the safe pattern. The `--components` flag accepts a comma-separated list.

What `migrate --components` actually accepts (main branch, read 10/05/2026). `--list-components` prints ten names: `dashboards`, `alerts`, `synthetics`, `slos`, `workloads`, `notification_channels`, `infrastructure`, `log_parsing`, `tags`, `drop_rules`. Of those, `migrate` exports and transforms only `dashboards`, `alerts`, `notification_channels`, `synthetics`, `slos` and `workloads`; any other name is reported as exported and produces nothing. Dependencies are added automatically — `alerts` pulls in `notification_channels`, and `slos` pulls in `alerts`. The table below maps NR surfaces to the engine's **transformers**; most of its first-column names are not `--components` values.

Component	Migrates	Transformer
<code>dashboards</code>	NR Dashboards → DT Documents (one per page), including funnel / honeycomb / event-feed / cascading variables / saved views	<code>dashboard_transformer</code> (Phase 11 + Phase 19 widget parity)
<code>alerts</code>	NR NRQL conditions → <code>builtin:davis.anomaly-detectors</code> + Workflow	<code>alert_transformer</code> (Phase 11 + Phase 17 extensions)
<code>baseline_alerts</code>	NR baseline / outlier NRQL conditions → Dynatrace Intelligence adaptive detectors	<code>baseline_alert_transformer</code> (Phase 17)
<code>non_nrql_alerts</code>	Infrastructure / Synthetic / Browser / Mobile / External-service / Multi-location-synthetic conditions	<code>non_nrql_alert_transformer</code> (Phase 17)
<code>lookup_tables</code>	NR <code>WHERE IN</code> lookups → Resource Store JSONL + DQL <code>lookup_subquery</code>	<code>lookup_table_transformer</code> (Phase 17)
<code>maintenance_windows</code>	NR scheduled + recurring maintenance + mute rules → DT Maintenance Windows + detector filters	<code>maintenance_window_transformer</code> (Phase 17)
<code>change_tracking</code>	NR deployment markers → DT events API (<code>CUSTOM_DEPLOYMENT</code> , <code>CUSTOM_CONFIGURATION</code>)	<code>change_tracking_transformer</code> (Phase 17)
<code>custom_event_ingest</code>	NR custom events → bizevent CloudEvent payloads	<code>custom_event_ingest_transformer</code> (Phase 17)
<code>synthetics</code>	NR Ping / Browser / Scripted API / Scripted Browser → <code>dynatrace_http_monitor</code> / <code>dynatrace_browser_monitor</code> (Config-v1, <code>ExternalSyntheticIntegration</code> scope)	<code>synthetic_transformer</code> (Phase 11)
<code>synthetics_specialized</code>	CERT_CHECK + BROKEN_LINKS	<code>synthetic_specialized_transformer</code> (Phase 24)
<code>slos</code>	NR SLOs (v1 / v2) → <code>builtin:monitoring.slo</code>	<code>slo_transformer</code> (Phase 11)
<code>key_transactions</code>	NR Key Transactions → SLO + OpenPipeline enrichment + Workflow bundle	<code>key_transaction_transformer</code> (Phase 23)
<code>workloads</code>	NR Workloads → Grail Filter Segment (<code>dynatrace_segment</code>) + bucket-scoped IAM	<code>workload_transformer</code> (Phase 11, Gen3 default)
<code>tags</code>	NR tag rules → OpenPipeline enrichment (<code>builtin:openpipeline.*</code>)	<code>tag_transformer</code> (Phase 11, Gen3 default)
<code>logs</code>	NR log forwarding configs → DT ingest or OneAgent log collection runbook	<code>log_parsing_transformer</code> + runbook (Phase 11)
<code>drops</code>	NR drop rules → OpenPipeline filter processors	<code>drop_rule_transformer</code> (Phase 11)
<code>parsing</code>	NR Grok parsing → OpenPipeline DPL <code>parse</code> processors	<code>log_parsing_transformer</code> (Phase 11)
<code>log_obfuscation</code>	NR PII / PAN masking → OpenPipeline <code>mask</code> processors (7 presets + regex)	<code>log_obfuscation_transformer</code> (Phase 17)

Component	Migrates	Transformer
log_archive	NR Log Live Archive → Grail bucket + OpenPipeline egress (S3/GCS/Azure Blob)	log_archive_transformer (Phase 24)
notifications	NR Notification Channels → Workflow action tasks (email/slack/pagerduty/webhook first-class; Jira/ServiceNow/OpsGenie/xMatters/VictorOps/Teams via http-function fallback or --legacy typed shapes)	folded into alert_transformer (Phase 11 + 17)
identity	NR Users / Teams / Roles / SAML → builtin:iam.* + SCIM runbook	identity_transformer (Phase 17)
cloud_integrations	AWS (16 services) / Azure (8 resources) / GCP (8 services)	cloud_integration_transformer (Phase 18)
kubernetes	NR K8s → DynaKube (full-stack / host-only) + Helm values	kubernetes_transformer (Phase 18)
aiops	NR AI Workflows + enrichments + decisions → DT automation + Dynatrace Intelligence detectors	aiops_transformer (Phase 18)
vulnerability	NR Vuln Mgmt → RVA alerting + per-CVE muting	vulnerability_transformer (Phase 18)
npm	SNMP devices + NetFlow collector (secrets redacted)	npm_transformer (Phase 18)
ai_monitoring	Model registry + inference NRQL → DQL mapping	ai_monitoring_transformer (Phase 18)
prometheus	NR Prometheus → DT scrape + OTLP remote-write + relabel filters	prometheus_transformer (Phase 18)
otel_metrics	NR OTLP metrics ingest → builtin:otel.ingest.metrics + collector YAML	otel_metrics_transformer (Phase 23)
otel_collector	NR OTEL collector (traces + metrics + logs + 5 processor kinds)	otel_collector_transformer (Phase 24)
statsd	NR StatsD → ActiveGate builtin:statsd.metrics + tag-mapping translation	statsd_transformer (Phase 23)
cloudwatch_metric_streams	NR Firehose path → builtin:aws.metric-streams + Terraform snippet	cloudwatch_metric_streams_transformer (Phase 23)
database_monitoring	10 DB engines (MySQL / Postgres / MSSQL / Oracle / MongoDB / Redis / Cassandra / MariaDB / DB2 / HANA) → builtin:dynatrace.extension.db.*	database_monitoring_transformer (Phase 24)
on_host_integrations	12 integrations (NGINX / HAProxy / Kafka / RabbitMQ / Elasticsearch / Memcached / Couchbase / Consul / Apache / etcd / Varnish / Zookeeper) → DT extensions	on_host_integration_transformer (Phase 24)
security_signals	NR Security Signals / IAST → AppSec envelope + per-signature OpenPipeline enrichment	security_signals_transformer (Phase 24)
custom_entity	NR custom entities → DT custom-device POST payload + enrichment matcher	custom_entity_transformer (Phase 24)
metric_normalization	NR rename / aggregate / drop rules → OpenPipeline metric processors	metric_normalization_transformer (Phase 24)
saved_filter_notebook	NR Data Apps → Document API type=='notebook' with markdown + DQL cells	saved_filter_notebook_transformer (Phase 24)

Recommended order: workloads first (foundations), then dashboards , then synthetics , then alerts (with notification_channels), then slo . Tag rules, log parsing, drop rules and the specialized surfaces in the table (cloud integrations, Kubernetes, Prometheus, database and on-host integrations, vulnerabilities, security signals, AI monitoring, OpenTelemetry, StatsD, CloudWatch metric streams, NPM, custom entities, metric normalization, saved-filter notebooks) are migrated by hand from the transformer guidance until migrate exports them.

5. Export Formats — Monaco & Terraform

Beyond direct API import, `Dynatrace-NewRelic` can emit **config-as-code** in two formats:

Monaco YAML

```
python3 migrate.py export-monaco --input ./output --output ./monaco-project
```

Output: a Monaco v2 project (`manifest.yaml` + per-config-type directories with `config.yaml` + JSON templates). Suitable for the SaaS Upgrade Assistant tar.gz upload pattern documented in the **M2S** (Managed-to-SaaS) series.

Terraform HCL

```
python3 migrate.py export-terraform --input ./output --output ./terraform-project
```

Output: HCL files using the `dynatrace-oss/dynatrace` provider. Resources include `dynatrace_document` (Gen3 dashboards), `dynatrace_davis_anomaly_detectors` (Gen3 detection) or `dynatrace_metric_events` (classic metric events), `dynatrace_slo_v2`, and the per-data-type `dynatrace_openpipeline_v2_<type>_*` family (e.g. `dynatrace_openpipeline_v2_logs_pipelines`), etc.

Why both formats

- **Monaco** is best for one-time bootstrap and bulk upload via SUA
- **Terraform** is best for ongoing GitOps-style management

The general guidance is **Terraform as the long-term tool**, with Monaco reserved for export and one-time deploys. The **AUTOM** series covers configuration automation tooling in depth.

6. End-to-End Runbook

This is the operational sequence to run a complete NR → DT migration.

Prerequisites

- [] NR API key + account ID configured
- [] DT Platform Token configured (OAuth / Classic tokens supported but Platform Token is the recommended default)
- [] Bucket strategy in place (see **ORGNZ-02 / ORGNZ-99**)
- [] Host group strategy in place (see **FAQ-01**)
- [] Stakeholders identified per wave

Phase 0 — Preflight (new in Phase 14; always do this first)

```
python3 migrate.py preflight
```

Probes the target tenant for Settings 2.0 / Document API / Automation API availability. If any Gen3 surface is missing, the command suggests `--legacy` and exits non-zero — catch this before burning a run.

Phase 1 — Discover (pair with NR2DT-01)

```
# Export from New Relic (nothing is written to Dynatrace)
python3 migrate.py migrate --export-only --output ./inventory
# Transform and write the conversion report without importing
python3 migrate.py migrate --dry-run --report --output ./inventory
```

Do not run `migrate --report` without `--dry-run`: with neither `--export-only` nor `--import-only`, `migrate` runs export, transform **and import**. Review `inventory/exports/newrelic_export.json`, enriched conversion-quality report (`confidence_score`, `warning_codes`, `runbook_url` per entry), and gap analysis. Stakeholder sign-off on wave plan.

Pre-decommission archive (Phase 17):

```
python3 migrate.py archive --account &lt;nr-account-id>; --since '30 days ago' \
--event-types Transaction,Log,SyntheticCheck --output ./nrdb-archive
```

Historical NRDB data is **not migratable to Grail** — archive as JSONL before decommissioning NR. Resumable per-event-type cursors let you interrupt and restart safely.

Phase 2 — Translate & Stage

```
# Translate NRQL inventory
# Extract NRQL from the export (produces inventory/all-nrql.txt + .csv)
python3 migrate.py extract-nrql --input ./inventory --output ./inventory/all-nrql.txt
python3 migrate.py extract-nrql --input ./inventory --output ./inventory/all-nrql.csv

# Translate — compile for plain DQL; batch for a CSV with confidence scores
python3 migrate.py compile --file inventory/all-nrql.txt --output translated.dql
python3 migrate.py batch --file inventory/all-nrql.csv --output translated.csv

# Instrumentation scan for newrelic.*() SDK calls (Phase 16) — one file per run
python3 migrate.py scan-instrumentation --file ./app-source/checkout/Handler.java --output ./instrumentation-todos.diff

# APM agent plan per language (Phase 16)
python3 migrate.py agents --language java --dry-run
python3 migrate.py agents --language nodejs --dry-run

# Transform each component and diff against the live tenant (nothing is imported)
python3 migrate.py migrate --dry-run --diff --report --components workloads --output ./wave0
```

Phase 3 — Wave-by-Wave Import (Gen3 default)

Use `migrate --dry-run --diff` before each wave and `--canary` for production tenants. Each wave below is a full run (export → transform → import), which is what writes `<output>/rollback-manifest.json`; `--diff` is ignored with `--import-only`.

```
# Wave 0 (foundations)
python3 migrate.py migrate --dry-run --diff --components workloads --output ./wave0
python3 migrate.py migrate --components workloads --canary 10 --canary-auto-proceed --output ./wave0

# Wave 1 (dashboards — read-only impact)
python3 migrate.py migrate --dry-run --diff --components dashboards --output ./wave1
python3 migrate.py migrate --components dashboards --output ./wave1

# Wave 2 (synthetics)
python3 migrate.py migrate --components synthetics --output ./wave2

# Wave 3 (alerts + notification channels — dual-alert window!)
python3 migrate.py migrate --components alerts,notification_channels --output ./wave3
# ... wait 1-2 weeks dual-alert ...

# Wave 4 (SLOs — also pulls in alerts and notification channels as dependencies)
python3 migrate.py migrate --components slos --output ./wave4
python3 migrate.py audit-slos

# Waves 5-6 (tags, logs, drop and parsing rules, specialized surfaces):
# not exported by `migrate` on the current main branch — migrate by hand (NRLC-07, §4)
```

Phase 4 — Validate & Cutover (NRLC-08)

Per-wave validation gates:

```
# Config-drift audit: live tenant vs the wave's transformed baseline (Phase 20)
python3 migrate.py audit --baseline ./wave1/transformed/dynatrace_config.json

# SLO validity audit (DT side only: SLOs evaluate, metrics exist)
python3 migrate.py audit-slos
```

Behavioral validation — the same query run in New Relic and in Dynatrace over the same window, results compared — has no CLI command; do it by hand per NRLC-08 §4. Sign-off per wave.

Phase 5 — Decommission NR

Disable NR alert routes; archive dashboards; halt NR ingest. Retain rollback manifests for 30+ days. `DynatraceClient.delete_entity` (Phase 20) now dispatches unified Gen3 deletes — rollback actually executes.

7. CI/CD Integration Patterns

Once the migration is run, ongoing config can live in CI:

GitOps with Terraform

```
# .github/workflows/dt-config.yml
name: dt-config
on: [push]
jobs:
  apply:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3
      - run: terraform init
      - run: terraform plan -out=tfplan
      - run: terraform apply tfplan
    env:
      DT_TOKEN: ${ secrets.DT_TOKEN }
```

Translation in CI

Run the compiler in CI to validate any committed DQL:

```
- run: npx nrql-translator validate --file ./queries.nrql
- run: |
    if [[ $(jq '.confidence' result.json) == "LOW" ]]; then
      echo "::error::LOW-confidence translation – manual review required"
      exit 1
    fi
```

Drift Detection

Schedule a daily drift audit to detect manual DT changes that drifted from the GitOps source of truth — Phase 20 `migrate.py audit` :

```
# audit exits 1 when it finds drift; the JSON report carries drift_count and by_kind
if ! python3 migrate.py audit --baseline ./output/transformed/dynatrace_config.json --output drift.json; then
  alert_team "DT config drift detected"
fi
```

Summary

The toolchain is open-source, modular, and CI-friendly. `Dynatrace-NewRelic` orchestrates; `nrql-engine` translates; `nrql-translator` exposes the engine on the CLI. Together they cover discovery, translation, transformation, validation, drift audit, canary promotion, and rollback. Pair them with the bucket strategy (**ORGNZ-02 / ORGNZ-99**) and host grouping strategy (**FAQ-01**) for a complete migration foundation.

Series Complete

You've reached the end of the NRLC deep-dive series. For the procedural step-by-step migration walkthrough, see the **NR2DT** series (9 procedural steps + NR2DT-99 summary). For reference, NRLC covers:

#	Topic
01	Platform Comparison
02	NRQL → DQL Translation
03	Dashboard Migration
04	Alert & Workflow Migration
05	Synthetic Monitor Migration
06	SLO & Workload Migration
07	Logs, Tags & Drop Rules
08	Validation, Diff & Rollback
09	Toolchain Reference & End-to-End Runbook

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